1️⃣ Regression Testing

✅ Definition (Based on Transcript + Industry Standard)

Regression Testing is performed to ensure that **new code changes (bug fixes, enhancements, new features)** have **not broken the existing functionality** of the application.

Whenever developers fix bugs or modify code, testers must verify that **all previously working modules still work correctly** and the application's behavior has not regressed.

📌 When Regression Testing is done?

- After bug fixes
- After enhancements
- After newly added features
- After code refactoring
- After environment upgrades

📌 Types of Regression Testing

1. **Unit Regression Testing**
   - Testing only the modified unit/module.
   - Other modules are NOT tested.
   - Done by developers.
2. **Partial Regression Testing**
   - Testing the changed module + its **impacted modules**.
   - Ensures that integration points still work.
3. **Full Regression Testing**
   - Testing the **entire application** when large changes are made.
   - Usually done before a major release.

🧪 Example

If a developer fixes an issue in "Purchase Module", regression ensures:

- Purchase module works after fix ✓
- Dependent modules like Inventory, Billing, Invoice also work ✓

⭐ Interview Importance

Regression testing is **one of the most frequently asked interview topics**.

❓ Different Ways This Question Is Asked in Interviews

- What is Regression Testing?
- Why is Regression Testing needed?
- When do you perform Regression Testing?
- What is the difference between full and partial regression?
- How do you select test cases for regression?
- Can automation be used for regression?
- How is regression testing different from re-testing?

2️⃣ Re-testing (Confirmation Testing)

✅ Definition

Re-Testing is performed to **verify whether the defect fixed by the developer is actually resolved**.

It is also called **Confirmation Testing**.

📌 Key Characteristics

- Always done on **failed test cases only**
- Always performed on the **same environment**
- It is done **before regression testing**
- Test cases are executed again **with the same input data**

🧪 Example

If Login fails due to incorrect password validation:

1. Tester reports the bug
2. Developer fixes it
3. Tester performs **Re-testing** for that specific login scenario

4. Then tester performs **Regression Testing** on surrounding functionality

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⭐ Interview Importance

Often asked with regression testing.

❓ Possible Interview Questions

- What is Re-testing?
- How is Re-testing different from Regression Testing?
- Do you perform Re-testing with same or different data?
- Is Re-testing mandatory for every bug?
- Can Re-testing be automated?

3️⃣ Difference Between Regression vs Re-testing

Feature	Re-testing	Regression Testing
Purpose	Check if **specific defect** is fixed	Check if **existing features** work after changes
Scope	Only failed test cases	Large set / full app
Data	Same data used earlier	Same or new data
Execution Priority	First	After Re-testing
Mandatory?	Yes	Based on changes
Automation	Hard to automate	Highly suitable for automation

⭐ Interview Importance

One of the **top 5 most commonly asked** testing questions.

❓ Interview Variants

- Differentiate Re-testing and Regression testing
- Examples of both

- Which comes first?
- Can regression be done without re-testing?
- Is regression testing mandatory?

4️⃣ Smoke Testing

✅ Definition

Smoke Testing is a **basic-level** testing performed on a new build to check whether the application is **stable enough** for further testing.

It checks **critical functionalities only**, such as:

- Login
- Navigation
- Basic UI loading
- Main workflows opening

📌 Purpose

To ensure:

- Build is stable
- Build is testable
- No show-stopper issues exist

📌 Performed By

- Testers (most common)
- Developers (in Dev environment)

🧪 Example

When QA receives a new build, they check:

- Login works
- Dashboard loads
- Pages open

If smoke fails → build is **rejected**.

⭐ Interview Relevance

Frequently asked in QA interviews.

❓ Possible Questions

- What is Smoke Testing?
- When do you perform Smoke Testing?
- Difference between Smoke vs Sanity testing
- Who performs smoke testing?
- Is smoke testing manual or automated?

5️⃣ Sanity Testing

✅ Definition

Sanity Testing is a **narrow and deep** testing performed after receiving a build with minor fixes or small changes to ensure that:

- The fixed issue works
- Related areas are not broken

It is a **subset of Regression Testing**.

📌 Key Characteristics

- Done when time is limited
- Focus on **specific functionalities**
- Build acceptance after minor fixes

⭐ Interview Questions

- What is Sanity Testing?
- When do we perform Sanity Testing?
- Why is it called a subset of regression?
- Difference between sanity & smoke?

6️⃣ Exploratory Testing

✅ Definition

Exploratory Testing is performed **without test cases** and **without detailed documentation**, where testers explore the application to understand functionality and identify defects.

📌 Characteristics

- Simultaneous learning, testing, and designing
- Useful when documentation is missing
- Relies on tester's domain experience
- Finds hidden defects
- Performed early in testing life cycle or in Agile

🧪 Example

You join a new project. Application is ready but no requirement document exists. You explore:

- All pages
- Features
- Links
- Validations
- Workflows

Then you document your findings → "Exploratory Notes".

⭐ Interview Variants

- What is Exploratory Testing?
- When do you use it?
- How is it different from Ad-hoc testing?
- What skills are required for exploratory testing?

7️⃣ Ad-hoc Testing

✅ Definition

Ad-hoc Testing is **informal, unstructured testing** performed without documentation, but with a specific intention:
👉 **Try to break the application by using random inputs.**

📌 Characteristics

- No documentation
- No planning
- Tester tries random workflows
- Goal: find unexpected defects

⭐ Interview Variants

- What is Ad-hoc testing?
- Difference between exploratory vs ad-hoc?
- Do we need experience for ad-hoc testing?
- How do testers perform it?

8️⃣ Monkey/Gorilla Testing

⚡ Monkey Testing

Random testing performed **without any knowledge** of the application.

⚡ Gorilla Testing

Testing **one module repeatedly** until it breaks.

🧪 Examples

- Monkey Testing: Clicking random buttons on a mobile app without knowing functionality.
- Gorilla Testing: Testing only “Login” repeatedly with 200+ inputs until it crashes.

⭐ Interview Variants

- What is monkey testing?
- What is gorilla testing?
- Difference between monkey, ad-hoc & exploratory
- When do you perform monkey testing?

9️⃣ Differences Between Exploratory, Ad-hoc, and Monkey Testing

Type	Knowledge of Application	Structured?	Purpose
Exploratory	Yes	Semi-structured	Learn → Test → Document
Ad-hoc	Some	Unstructured	Break the system
Monkey	None	Completely random	Stress system using random inputs

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⭐ Interview Importance

Common question in mid-level interviews.

❓ Variants

- Difference among ad-hoc, monkey, exploratory
- Give real project examples
- When to use which technique?

🔟 Positive & Negative Testing

🔹 Positive Testing

Verifying the application behaves **as expected** with **valid inputs**.

🔹 Negative Testing

Verifying the application behaves **correctly** with **invalid inputs**, unexpected inputs, or wrong workflows.

📌 **Example – Login Page**

- Positive: Correct username + correct password → Login success
- Negative:
  - Wrong username
  - Empty password
  - Special characters

⭐ **Interview Variants**

- What is positive vs negative testing?
- Why do we need negative testing?
- Examples from your project?
- How do you generate negative scenarios?

1️⃣1️⃣ End-to-End (E2E) Testing

✅ Definition

Testing the **complete workflow** of the application from start to finish, covering all integrated systems.

🧪 Example – E-commerce

1. Login
2. Search Product
3. Add to Cart
4. Checkout
5. Payment
6. Order Confirmation

⭐ **Interview Variants**

- What is E2E testing?
- Difference between system testing and E2E testing?

- Example of E2E testing from your project?

1️⃣2️⃣ Localization Testing

✅ Definition

Testing how well the application adapts to a **specific locale**, including:

- Language
- Currency
- Date formats
- Local laws
- Number formats

Example

English → Hindi
USD → INR
MM/DD/YYYY → DD/MM/YYYY

1️⃣3️⃣ Internationalization / Globalization Testing

📝 Internationalization (i18n)

Designing software so it can be easily localized.

📝 Globalization

Ensuring the product works across multiple regions, languages, and cultures.

❓ Interview Variants

- What is localization testing?
- What is internationalization testing?
- i18n vs L10n
- Real examples from your project?